SMART LIBRARY REQUEST FLOW

Abstract :

>The Smart Library Request Workflow in ServiceNow helps manage book requests easily. A student requests a book, and the request is sent to the librarian using ServiceNow Flow Designer. The

librarian then approves or rejects the request. If approved, the book is issued and later returned with proper tracking. The system also generates reports to show the most borrowed books and request details.

Introduction :

- Library is used to manage books and users.
- Many libraries still use manual work, which takes more time.
- So, a Smart Library system is created to make the process easy and automatic.

Problem Statement:

>Library system is mostly manual and takes more time.

>It is difficult to track available and borrowed books.

>Book requests are not managed properly.

>Students cannot easily check book availability.

>Librarians face difficulty in approving and managing requests.

>There is no proper system for tracking and reports.

>So, a Smart Library system is needed to make the process easy, fast, and accurate.

Objectives:

- **Define roles (student, librarian) to enforce controlled access.**
- **Create and configure custom tables for catalog and borrow requests.**
- **Automate approvals and book issuance using Flow Designer.**
- **Apply UI Policies and ACLs to maintain consistency and security.**
- **Provide reporting capabilities for administrators to track library usage.**

Literature Review:

- **Many libraries use manual systems which take more time and effort.**
- **Some systems are digital but do not have proper automation.**
- **They lack features like easy request handling and tracking.**
- **There is also no proper report generation in some systems.**
- **So, a better system is needed with automation and easy management.**

Methodology:

- **Designed system using ServiceNow platform**
- **Created Book and Borrow Request tables**
- **Defined roles (Student, Librarian)**
- **Planned workflow using Flow Designer**
- **Applied access control (ACLs)**

Description:

Roles :

Student and librarian roles ,manage access permissions on request of books.

Custom Tables :

Book and Borrow Request tables store catalog and transaction data.

Reference Qualifier :

Ensures only available books can be requested.

Flow Designer :

Automates approval workflows and updates book

status.

UI Policies:

Lock or enforce field behavior after approval.

Access Control Rules (ACLs):

- Enforce role-based permissions for read,write, and delete operations.

Reports:

-Generate insights such as Most Borrowed Books.

PROJECT PHASES –

1. Create Roles:

- To create a role ,Provide a unique name and a clear description explaining the purpose of the role, then save the record.

Creation:

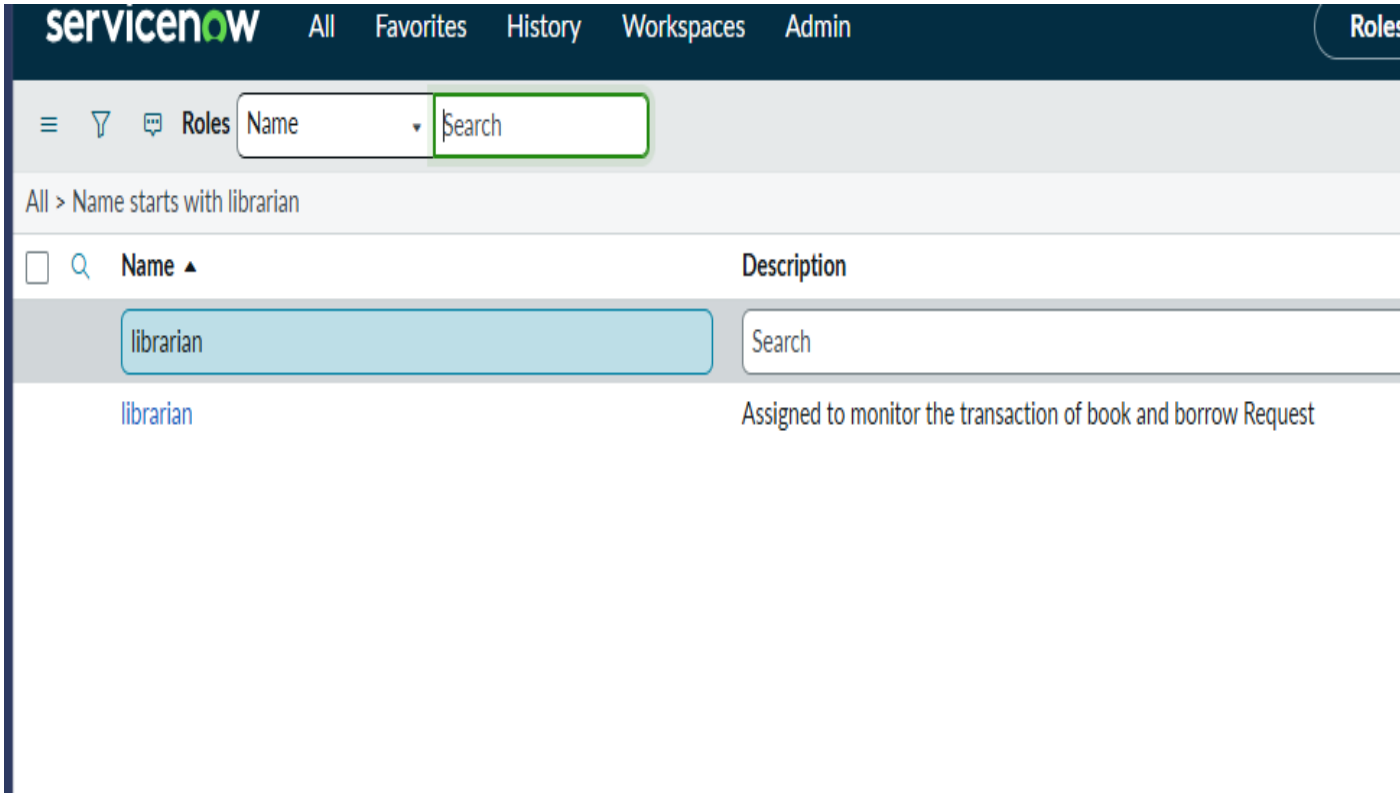
- Navigate to User Administration -> Roles.

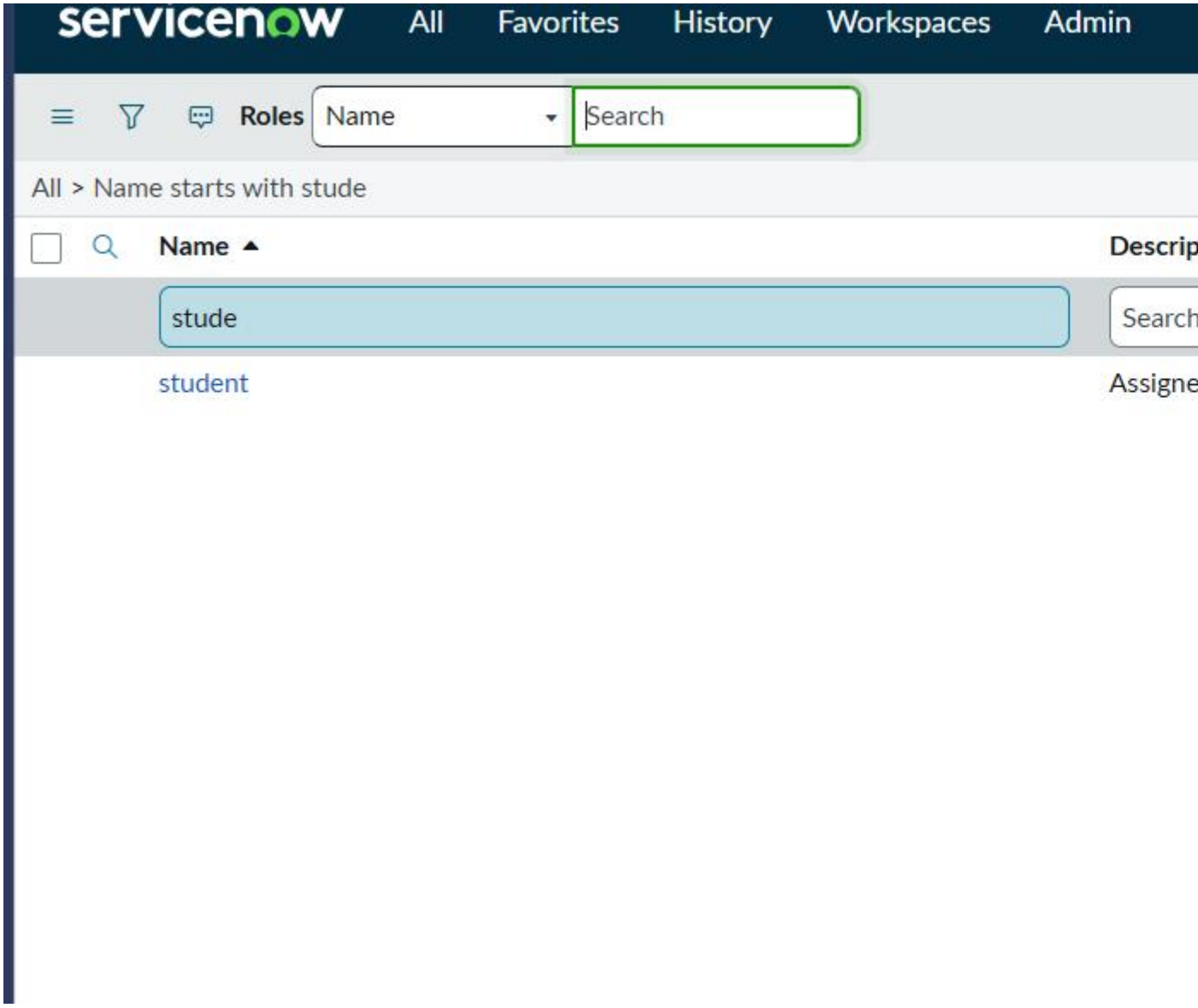
- **Create roles: librarian and student.**

- **Assign roles to users as required.**

ROLES CREATED:

Librarian and student




servicenow
All
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stude
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2.Table Creation :

- Navigate to System Definition -> Tables.
- Create tables: Book and Borrow Request.
- Configure fields such as Title, Author, Status, and Return Date.

TABLE CREATED:

Book and Borrow Request

Table
Book

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use table

* Label: Book
* Name: u_book

Application: Glob

Remote Table

Columns | Controls | Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length	
×	ISBN	String	(empty)		20
×	Title	String	(empty)		100
×	Status	Choice	(empty)		40
×	Author	String	(empty)		100
	Sys ID	Sys ID (GUID)	(empty)		32
	Created by	String	(empty)		40
	Created	Date/Time	(empty)		40
	Updates	Integer	(empty)		40
	Updated by	String	(empty)		40
	Updated	Date/Time	(empty)		40
×	Student	Reference	Borrow Request		32
+	*Insert a new row...*				

Delete | Update | Delete All Records

servicenow All Favorites History Workspaces Admin Table - Borrow Request

Table
Borrow Request

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applicatio

* Label Borrow Request

Applica

* Name u_borrow_request

Remote T

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

	Column label	Type	Reference	Max length
X	Status	Choice	(empty)	
X	Return Date	String	(empty)	
X	Book	Reference	Book	
	Created	Date/Time	(empty)	
	Updates	Integer	(empty)	
	Updated	Date/Time	(empty)	
	Created by	String	(empty)	
	Updated by	String	(empty)	
	Sys ID	Sys ID (GUID)	(empty)	
+	*Insert a new row...*			

Delete Update Delete All Records

Related Links

3.Configuring Table :

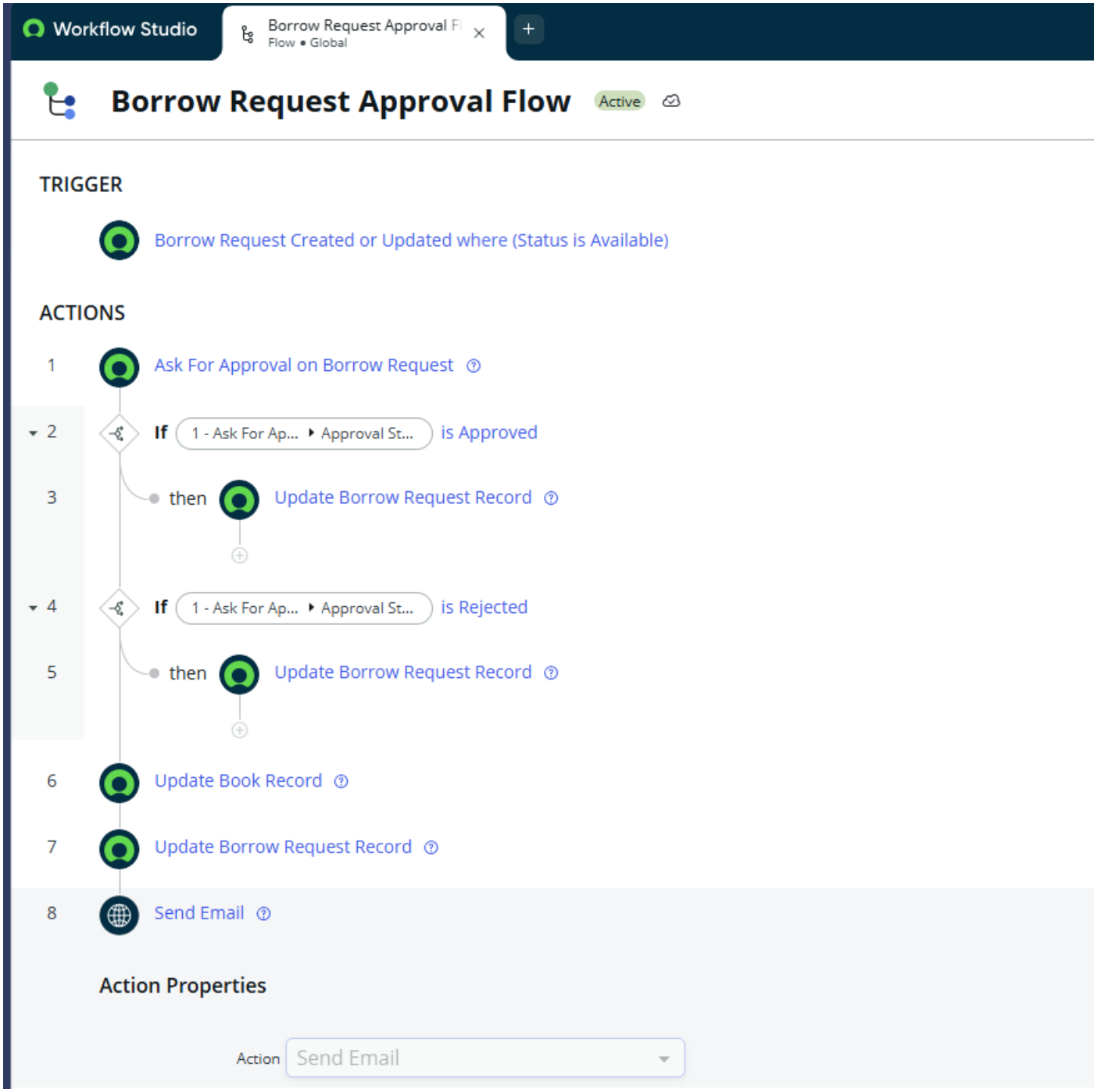
- Configure the Borrow Request table's Book field to reference only records where Status = Available.

4.Creating Flow Designer :

- Create a flow named Borrow Request Approval Flow.
- Trigger: Borrow Request created with Status = Requested - Actions.
- Approval by librarian.
- On approval, update Book.

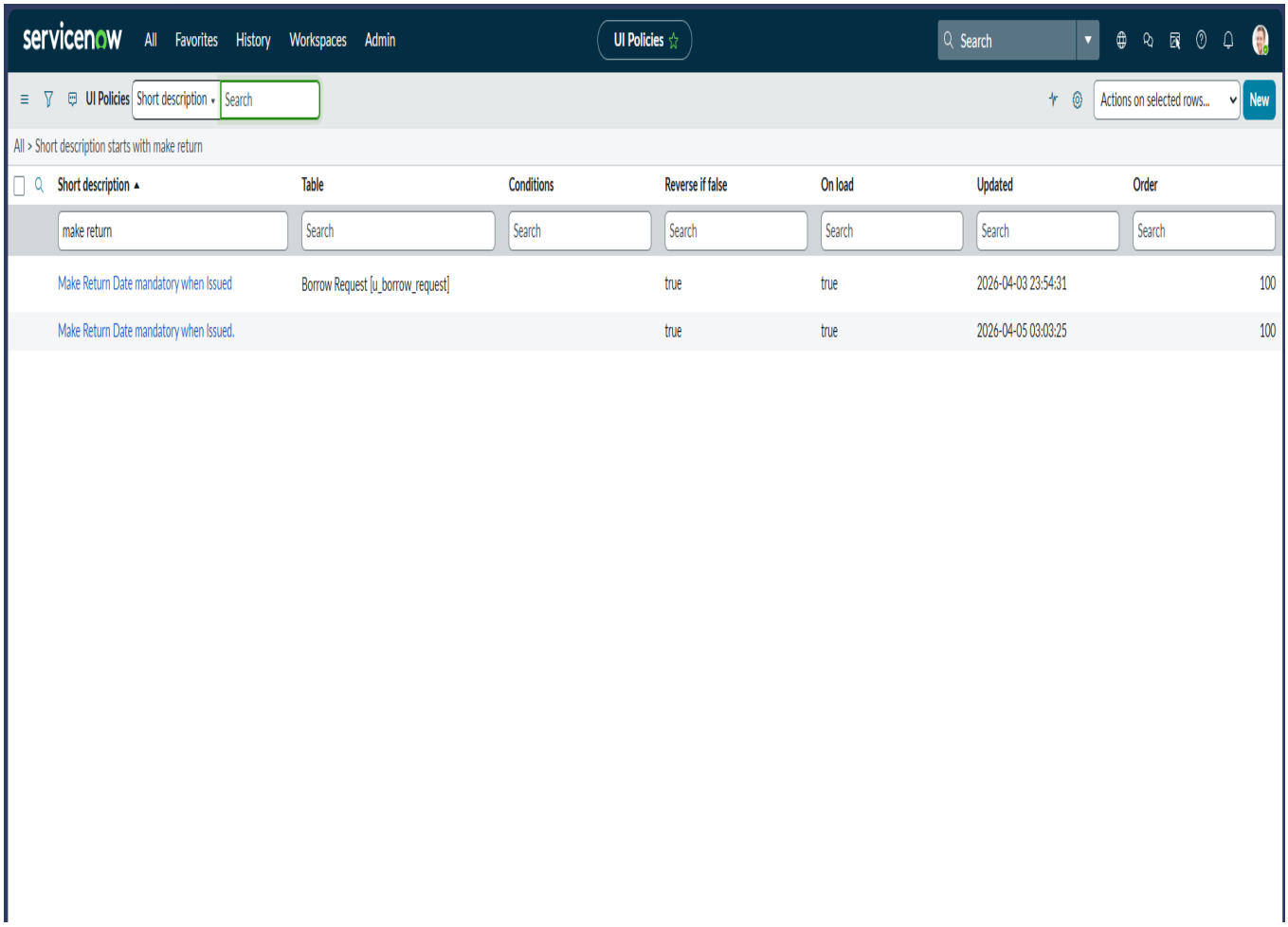
Status = Issued and Borrow Request.Status = Approved.

- Send email notification to student.



5. Creating UI Policies:

- Create a UI Policy for Borrow Request.
- Example: Make Return Date mandatory when Status = Issued.

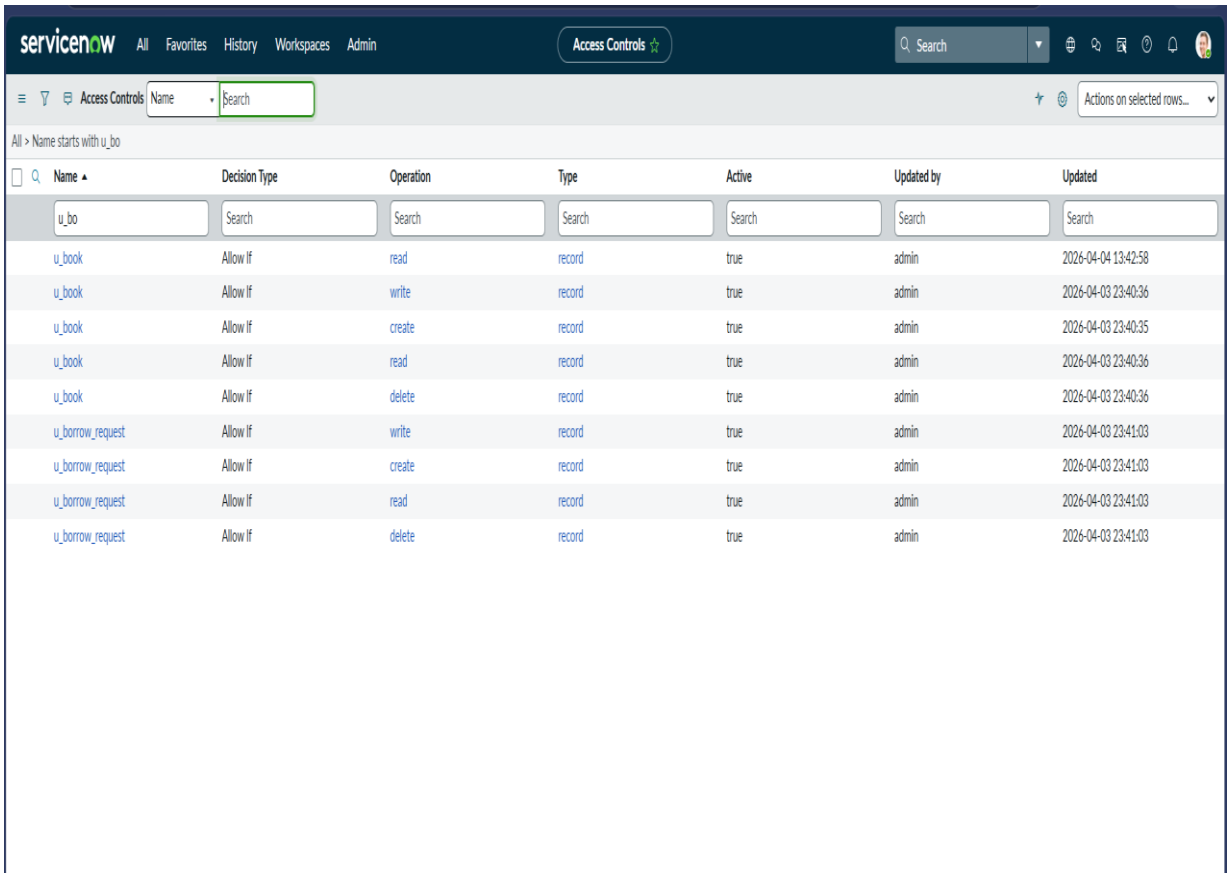


6. Creating Access Control List:

- Configure ACLs for Book table.

- **Students: read only.**

- **Librarians: read, write, delete.**



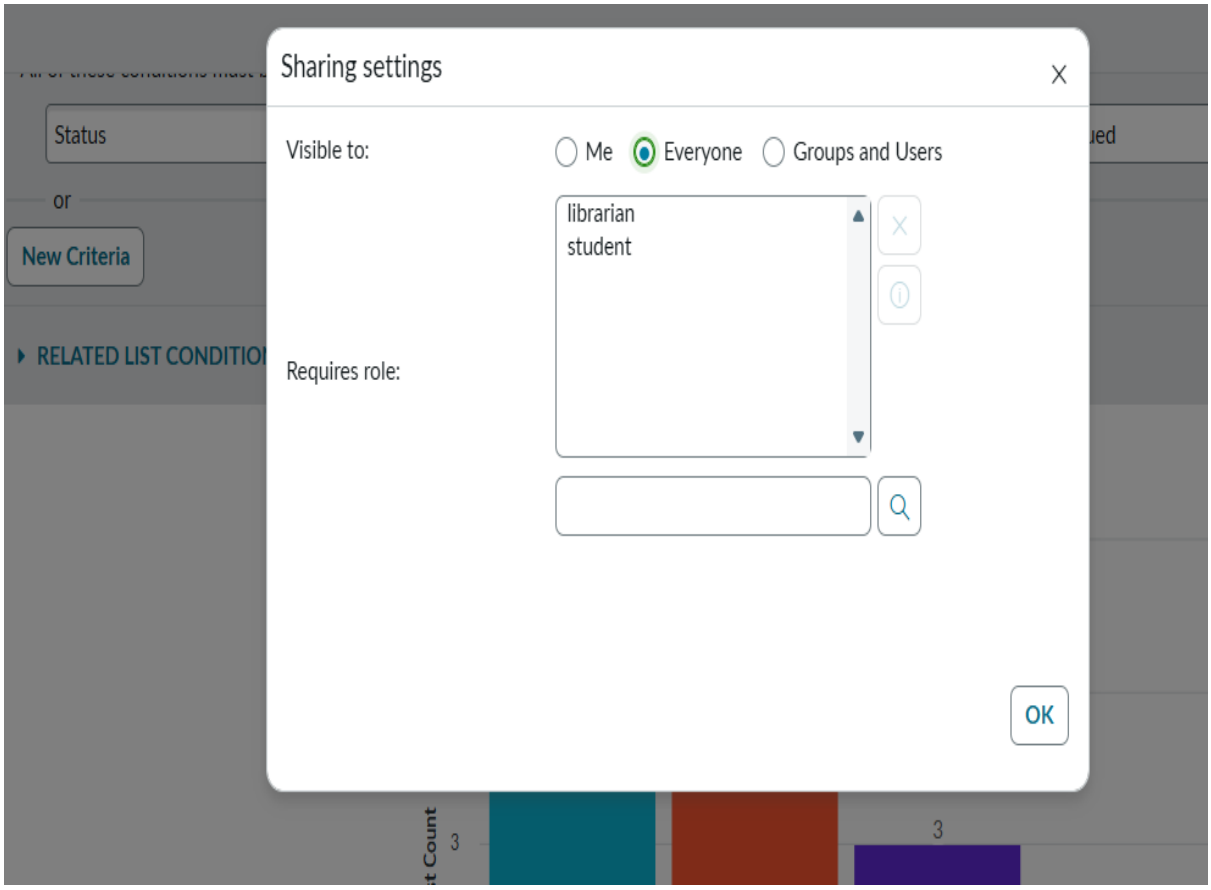
Name	Decision Type	Operation	Type	Active	Updated by	Updated
u_bo	Search	Search	Search	Search	Search	Search
u_book	Allow If	read	record	true	admin	2026-04-04 13:42:58
u_book	Allow If	write	record	true	admin	2026-04-03 23:40:36
u_book	Allow If	create	record	true	admin	2026-04-03 23:40:35
u_book	Allow If	read	record	true	admin	2026-04-03 23:40:36
u_book	Allow If	delete	record	true	admin	2026-04-03 23:40:36
u_borrow_request	Allow If	write	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	create	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	read	record	true	admin	2026-04-03 23:41:03
u_borrow_request	Allow If	delete	record	true	admin	2026-04-03 23:41:03

7. Create Report :

- **Create a report named Most Borrowed Books.**

- **Source: Borrow Request table.**

- **Group by Book, aggregate by Count, filter Status = Approved.**



Project Explanation:

- **The project integrates multiple ServiceNow features to build a functional**

- library management system. Roles ensure secure access, tables store
- structured data, and reference qualifiers maintain integrity. Flow
- Designer automates approvals and notifications, while UI Policies and ACLs enforce consistency and security. Reporting provides administrators with valuable insights into library usage patterns.

Implementation :

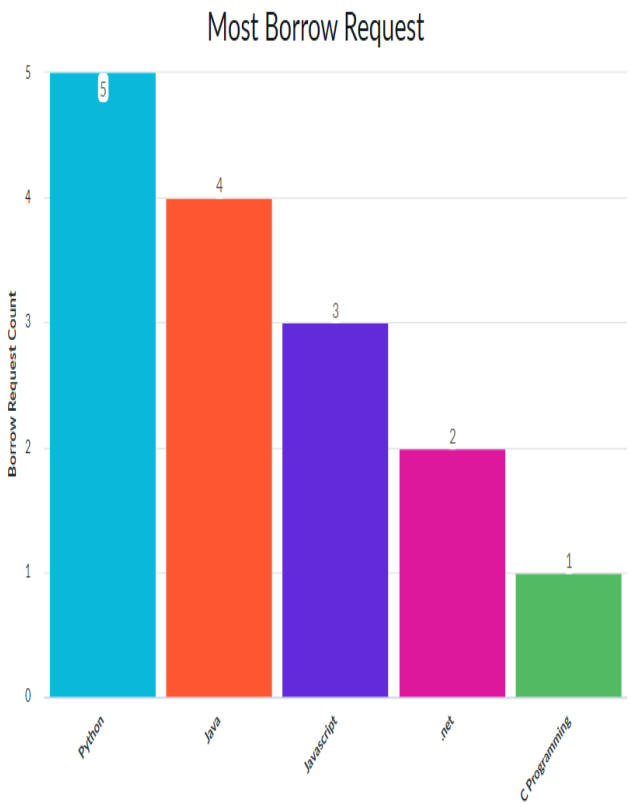
We built a Smart Library Request Workflow in ServiceNow. Students can request books online, and librarians can approve or reject requests. The system automatically tracks issued and returned books.

Results :

The workflow reduced manual work and improved request tracking. Book requests are processed faster, and records are maintained properly. Reports help identify the most borrowed books.

Table: Borrow Request [u_borrow_request]

All > Status = Issued



Conclusion:

The Library Management Project successfully optimizes book borrowing and management processes in ServiceNow. By combining automation, role-based

access, and reporting, the system enhances efficiency,

accountability, and transparency. Students benefit from a simplified

request process, while librarians gain streamlined approval workflows.

Future Scopes:

- Scheduled Jobs: Automate recurring tasks such as overdue reminders.

- Integration with Digital Resources: Extend to e-books and online

catalogs.

- Enhanced Notifications: SMS or mobile app alerts for approvals and

returns.

- Advanced Analytics: Track borrowing trends, user activity, and resource

utilization.

- Scalability: Extend framework to other academic services such as

reservations or asset tracking.

References :

Sources used ServiceNow official documentation, Flow Designer guides, and online tutorials related to IT Service Management (ITSM) and workflow automation.